

Scheduler Assistant: An Autonomous Agentic Calendar That Reasons

Aryaan Peshoton, Karan Vohra, Jamal Moussa

Problem

Current calendars cannot resolve conflicts or prioritize meetings.

LLMs lack persistent calendar state and constraint reasoning.

Our RRL-Scheduler combines calendar knowledge with reasoning to automatically schedule meetings.

Why it Matters

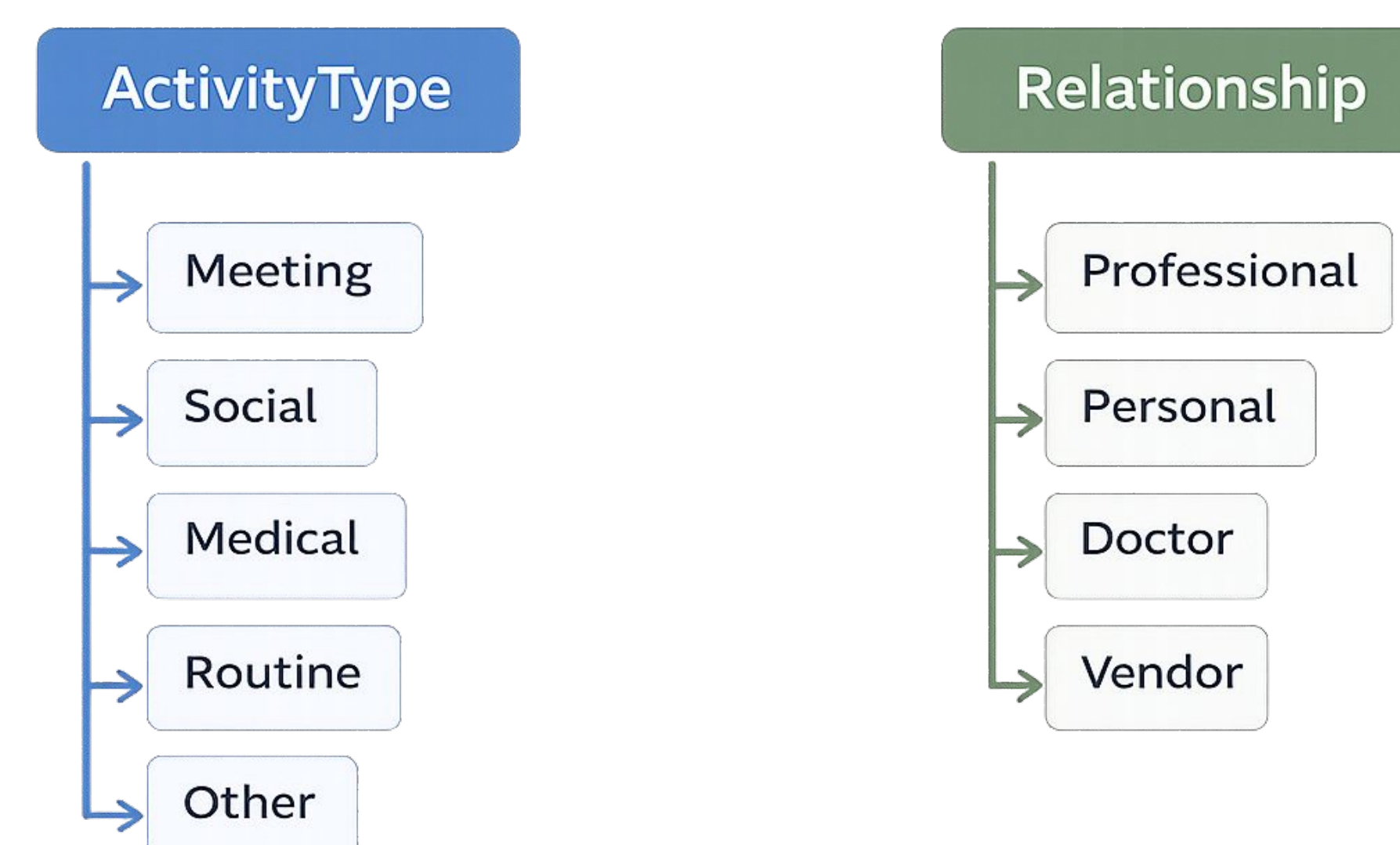
Reduces the time spent coordinating meetings by automatically resolving conflicts.

Prioritizes requests based on context.

Learns user preferences and adjusts decisions based on past behavior.

Representation

Knowledge Graph:



Persistent Stores (JSON):

Calendar, Contacts, Preferences, Memory, Requests, Routines

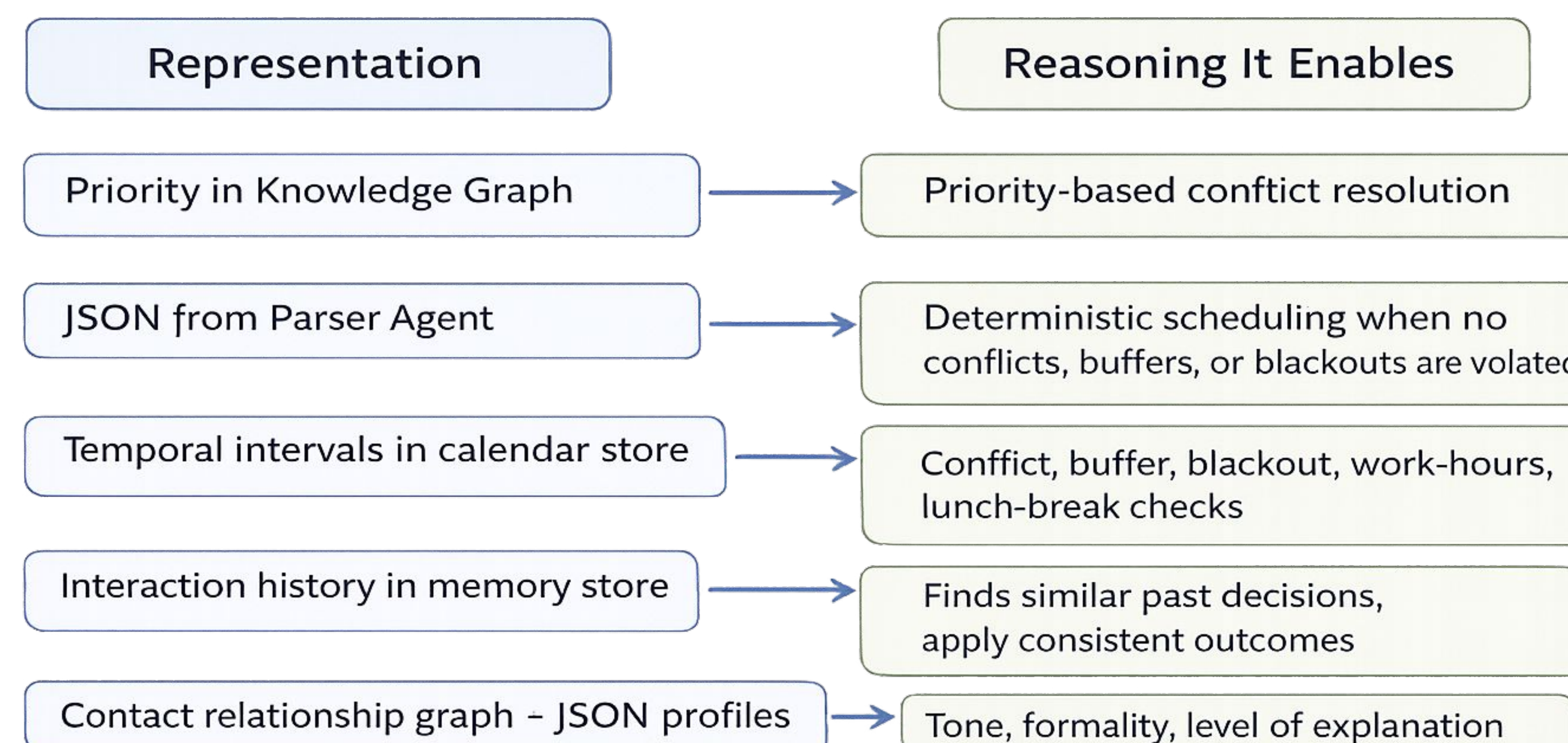
JSON Schemas:

- Structured inter-agent
- Messages
- Parsed intents
- Pending decisions

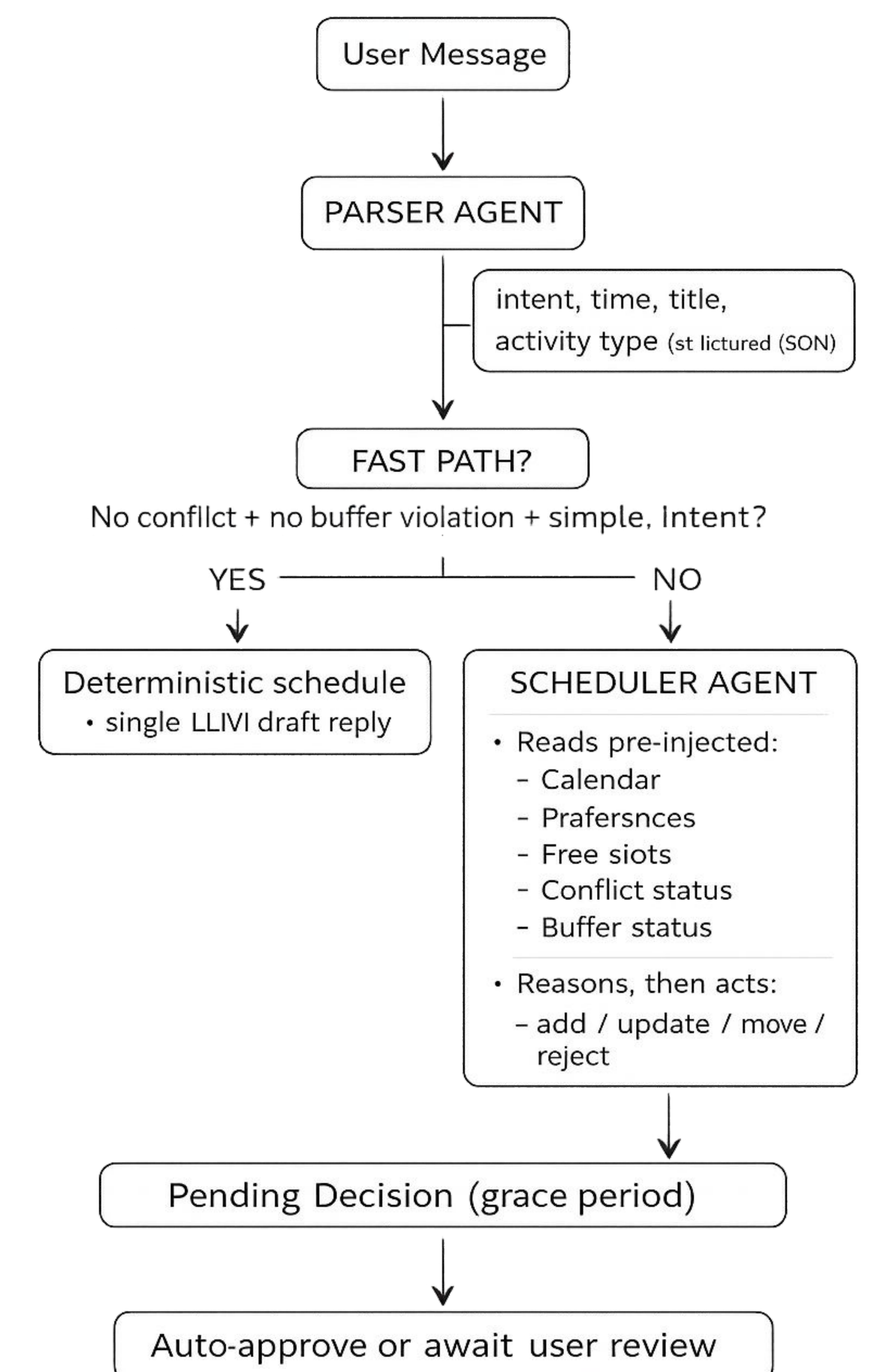
Tabular Views:

- Appointment rows
- Free-slot lists
- Preference summaries

Reasoning



Methods and How Reasoning Works



Reasoning Mechanisms:

Constraint Satisfaction
Priority Resolution
Proactive Rescheduling
Buffer Awareness
Case-Based Reasoning